What did you learn yesterday?

how does the following functions work in list

```
    a. append()
    b. extend()
    c. insert()

l = ["avocado", "banana", "lemon"]
l.append("orange")
print(l)
```

```
['avocado', 'banana', 'lemon', 'orange']
```

```
list1 = [1 ,4, 4]
list2 = [3, 90, 2]
list1.append(list2)
list1
```

```
[1, 4, 4, [3, 90, 2]]
```

```
list1 = [1 ,4, 4]
list2 = [3, 90, 2]
# list1.append(list2)
list1.extend(list2)
list1
```

```
[1, 4, 4, 3, 90, 2]
```

```
# list2


#make the list like: [3, 12, 90, 2]
list2.insert(1,12)
list2
```

```
[3, 12, 90, 2]
```

```
'''
If we have a string, s = "Amazingly" what will be the output of the
following:
    a. s[3] =>
    b. s[2:] =>
    c. s[::-2] =>
    d. s[:7:3] =>

'''
```

```
Given a list named "oneToFifty" with numbers from 1 to 50 [1, 2, 3, 4,
..... 50]
```

```python
oneToFifty = [i for i in range(1, 51)]
print(oneToFifty)
```

```
[1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19,
20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36,
37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50]
```

```
oneToFifty
```

```
[1,
 2,
 3,
 4,
 5,
 6,
 7,
 8,
 9,
 10,
 11,
 12,
 13,
 14,
 15,
 16,
 17,
 18,
 19,
 20,
 21,
 22,
 23,
 24,
 25,
 26,
 27,
 28,
 29,
 30,
 31,
 32,
 33,
 34,
 35,
 36,
 37,
 38,
 39,
 40,
 41,
```

```
 42,
 43,
 44,
 45,
 46,
 47,
 48,
 49,
 50]
```

```
#1. print multiples of 10: hint: use slicing Slicing: [start:end:step]
#oneToFifty = [1, 2, 3,4, 5, 6, 7, 8, 9, 10..... 50]
#1.1: in their natural order [10, 20, 30, 40, 50]
print(oneToFifty[9::10])
#1.2: in reverse order [50, 40, 30, 20, 10]
print(oneToFifty[-1:8:-10])
```

```
[10, 20, 30, 40, 50]
[50, 40, 30, 20, 10]
```

```
#multiples of two [2, 4, 6, 8....50]
```

```
[2,
 4,
 6,
 8,
 10,
 12,
 14,
 16,
 18,
 20,
 22,
 24,
 26,
 28,
 30,
 32,
 34,
 36,
 38,
 40,
 42,
 44,
 46,
 48,
 50]
```

```
#2. Reverse the list: [50, 49....2, 1]
```

```
[50,
 49,
 48,
 47,
 46,
 45,
 44,
 43,
 42,
 41,
 40,
 39,
 38,
 37,
 36,
 35,
 34,
 33,
 32,
 31,
 30,
 29,
 28,
 27,
 26,
 25,
 24,
 23,
 22,
 21,
 20,
 19,
 18,
 17,
 16,
 15,
 14,
 13,
 12,
 11,
 10,
 9,
 8,
 7,
 6,
 5,
 4,
 3,
```

```
 2,
 1]

#3. what is the index of the number 43

#4. how can we find the index of 43 imagine you don't know the order
```

1. Control flow

This is condition syntax

```
if CONDITIONAL:
    IF_BODY
elif CONDITIONAL: #optional
    ELIF_BODY
else: #optional
    ELSE_BODY

if condition: #the condition must be true to execute the ifblock
  .......



#Example of if statment
if True:
    print('Hello World!')

Hello World!

if False:
    print('Hello, World!')

a = 33
b = 200
if b > a: #True
  print("b is greater than a")

b is greater than a

# with conditional
if 1 == 2: #False
    print('Hello World!')

# with else condition
if 1 == 1: #False
    print('Hello World!')
else:
    print('Goodbye world!')

Hello World!
```

```
# Else is a catch-all
if 1 == 2:
    print('Hello World!')
else:
    print('Goodbye world!')   #Else is executed when the 'if'
condition is false
```

```
Goodbye world!
```

```
# elif only triggers on the condition
x = 1
if x == 0: #false
    print('x=0')
elif x == 2: #false
    print('x=2')
else:
    print("x is neither 0 nor 2")
```

```
x is neither 0 nor 2
```

```
# why are you guys quiet
'''
1. question = "yes/no"
2. scared = "yes/no"
3. forced = "yes/no"
'''

question = input("do you have a question?(yes/no)")
scared = input("ARE YOU SCARED? (yes/no)")
forced = input("Are you forced to come here? (yes/no)")
```

```
do you have a question?(yes/no)no
ARE YOU SCARED? (yes/no)no
Are you forced to come here? (yes/no)yes
```

```
if question == "yes": #false
  print("Please ask your question!")
elif scared == "yes": #false
  print("Please don't be scared!")
elif forced == "yes":#true
  print("Please try to enjoy!")
else:
  print("Enjoy!")
```

```
Please try to enjoy!
```

```
# we can combile both elif and else together
x = 1
if x == 0:
    print('x=0')
elif x == 2:
```

```
    print('x=2')
elif x == 3:
    print("x=3")
else:
    print('x is neither 0, 3 nor 2')
```

```
x is neither 0, 3 nor 2
```

```
x = 5
if x > 6: #false
    print("greater than 6")
elif x < 10: #true
    print("less than 10")
elif x < 6:
    print("less than 6")
else:
    print("I dunno")
```

```
less than 10
```

```
                if something (e.g. x == 0)
                            |
                            |
                       /---------\
                      /           \
                   True          False
                    /               \
                   /                 \
                  /                   \
            Do something           if something (e.g. x == 2) (elif)
         (e.g. print('x=0'))                |
                                            |
                                        ---------
                                       /         \
                                      /           \
                                    True         False
                                     /             \
                                    /               \
                                Do something      Do something (else)
                        (e.g. print('x=2'))      (e.g. print('x is
neight 0 or 2'))
```

Comparison operators:

- < less than
- > greater than
- >= greater than or equal to
- <= less than or equal to
- == equal to
- != not equal

```
1 != 2
```

```
True
```

```
grade = float(input("Insert your grade please: "))
# we have grade value here, if the grade is greater than or equal to
50 it will be pass and if it is less, Fail

if grade >= 50:
  print("PASS")
else:
  print("FAIL")
```

```
Insert your grade please: 98
PASS
```

'and' & 'or' operators

```
True and False
```

```
False
```

```
# Unity park ticket discount
# be above 15 years old and be student
#both of them should be true

age = 10
status = "student"

if age > 15 and status == 'student': #False and True = False
  print("get discount!")
else:
  print("no discount")
```

```
no discount
```

```
# we  can use 'and' and 'or' keywords in checking conditions
name = "John"
age = 23
if name == "John" and age == 23: #True and True = True
    print("Your name is John, and you are also 23 years old.")

if name == "John" or name == "Rick": #True or False = False
    print("Your name is either John or Rick.")
```

```
Your name is John, and you are also 23 years old.
Your name is either John or Rick.
```

```
# if #condition:
#   #body
```

```
# elif #condition:
#   #body
# elif #condition:
#   #body
# else:
#   #body
```

Grades: Math = 90, PHysics = 86, chemistyr = 76 Math = A, Physics = B, chemistry = D

```
#ask the user's grade
# >90: A
# 85 - 90: B
# 80 - 85: C
# 70 - 80: D
# <70: F

grade = float(input("please insert your grade? "))
if grade >= 90:
  print("A")
elif grade >= 85 and grade < 90:
  print("B")
elif grade >= 80 and grade < 85:
  print("C")
elif grade >= 70 and grade < 80:
  print("D")
else:
  print("F")

please insert your grade? 78
D
```

'in' operator

```
name = "Efrem"
items = ["John", "Rick", "Maereg", "Samson", "Mihretu", "Efrem"]
name in items

True

# using keyword 'in', it will check if the item is in the list or no
name = "John"
names = ["John", "Rick"]
if name in names: #True
    print("Your name is either John or Rick.")

Your name is either John or Rick.

l = ["Hello", 2, 4, "world"]
value  = 5
if value in l: #False
```

```
    print("the list has 5 in it")
else:
    print("5 is not in the list")
```

```
5 is not in the list
```

'not' operator

```
name = "Michael"
names = ["John", "Rick"]
if name not in names: #True
    print("Your name is neither John nor Rick.")
```

```
Your name is neither John nor Rick.
```

```
not False
```

```
True
```

Nested if

```
k = 10
print(k)
```

```
10
```

```
# You can have if statements inside if statements, this is called
nested if statements.
x = 41

if x > 10: #true
  print("Above ten,")
  if x > 20:
    print("and also above 20!")
  else:
    print("but not above 20.")
```

```
Above ten,
and also above 20!
```

```
# Unity park ticket discount
# be above 15 years old and be student
#both of them should be true

age = 10
status = "student"

if age > 15:
  if status == "student":
    print("get discount")
```

```
  else:
    print("no discount")

if age > 15 and status == "student":
  print("get discount")
else:
  print("no discount")
```

Exercise on if conditions

```
'''
    1. Take two int values from user and print greatest among them.
'''
num1 = int(input("enter the first number? "))
num2 = int(input("enter the second number? "))
if num1 > num2:
  print(num1)
else:
  print(num2)

#The greatest number fomr num1 and num2 is ____

enter the first number? 590
enter the second number? 900
900

'''
    1. Take three int values from user and print greatest among them.
    x, y, z
    x > y... x > z
    y > x, y > z..
    else: z
'''
x = int(input("enter the first number? "))
y = int(input("enter the second number? "))
z = int(input("enter the third number? "))

if x > y and x > z:
  print(f"the largest number is {x}")
elif y > x and y > z:
  print(f"the largest number is {y}")
else:
  print(f"the largest number is {z}")



enter the first number? 6
enter the second number? 9
enter the third number? 1
the largest number is 9
```

```
x = int(input("enter the first number? "))
y = int(input("enter the second number? "))
z = int(input("enter the third number? "))

if x > y:
  if x > z:
    print(f"the largest number is {x}")
elif y > x:
  if y > z:
    print(f"the largest number is {y}")
else:
  print(f"the largest number is {z}")
```

```
enter the first number? 7
enter the second number? 9
enter the third number? 1
the largest number is 9
```

```
11#let's ask user their gender in terms of M or F then write full
gender
gender = input("please insert your gender(M/F): ")

#write your if condition here
```